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Emerging and Legacy Perfluoroalkyl Substances in Breastfed Chinese Infants: Renal Clearance, Body Burden, and Implications

Jingzhi Yao,[1,2] Zhaomin Dong,[3] Lulin Jiang,[1] Yitao Pan,[1,4] Meirong Zhao,[5] Xiaoxia Bai,[6] and Jiayin Dai[1,4]

[1]State Environmental Protection Key Laboratory of Environmental Health Impact Assessment of Emerging Contaminants, School of Environmental Science and Engineering, Shanghai Jiao Tong University, Shanghai, China
[2]Key Laboratory of Animal Ecology and Conservation Biology, Institute of Zoology, Chinese Academy of Sciences, Beijing, China
[3]School of Space and Environment and Beijing Advanced Innovation Center for Big Data-Based Precision Medicine, Beihang University, Beijing, China
[4]State Key Laboratory of Reproductive Medicine, Center for Global Health, School of Public Health, Nanjing Medical University, Nanjing, China
[5]College of Environment, Zhejiang University of Technology, Hangzhou, Zhejiang, China
[6]Department of Obstetrics, Women's Hospital, School of Medicine, Zhejiang University, Hangzhou, Zhejiang, China

Background: Human breast milk is a primary route of exposure to perfluoroalkyl substances (PFAS) in infants. To understand the associated risks, the occurrence of PFAS in human milk and the toxicokinetics of PFAS in infants need to be addressed.

Objectives: We determined levels of emerging and legacy PFAS in human milk and urine samples from Chinese breastfed infants, estimated renal clearance, and predicted infant serum PFAS levels.

Methods: In total, human milk samples were collected from 1,151 lactating mothers in 21 cities in China. In addition, 80 paired infant cord blood and urine samples were obtained from two cities. Nine emerging PFAS and 13 legacy PFAS were analyzed in the samples using ultra high-performance liquid chromatography tandem mass spectrometry. Renal clearance rates (CL_{renal}s) of PFAS were estimated in the paired samples. PFAS serum concentrations in infants (<1 year of age) were predicted using a first-order pharmacokinetic model.

Results: All nine emerging PFAS were detected in human milk, with the detection rates of 6:2 Cl-PFESA, PFMOAA, and PFO5DoDA all exceeding 70%. The level of 6:2 Cl-PFESA in human milk (median concentration = 13.6 ng/L) ranked third after PFOA (336 ng/L) and PFOS (49.7 ng/L). The estimated daily intake (EDI) values of PFOA and PFOS exceeded the reference dose (RfD) of 20 ng/kg BW per day recommended by the U.S. Environmental Protection Agency in 78% and 17% of breastfed infant samples, respectively. 6:2 Cl-PFESA had the lowest infant CL_{renal} (0.009 mL/kg BW per day), corresponding to the longest estimated half-life of 49 y. The average half-lives of PFMOAA, PFO2HxA, and PFO3OA were 0.221, 0.075, and 0.304 y, respectively. The CL_{renal}s of PFOA, PFNA, and PFDA were slower in infants than in adults.

Conclusions: Our results demonstrate the widespread occurrence of emerging PFAS in human milk in China. The relatively high EDIs and half-lives of emerging PFAS suggest potential health risks of postnatal exposure in newborns. https://doi.org/10.1289/EHP11403

Introduction

Per- and polyfluoroalkyl substances (PFAS) are a broad range of synthetic chemicals widely used as water- and oil-repellents in textiles, food packaging, and leather; processing aids in the fluoropolymer production; anti-mist agents in chrome plating; and surfactants in film-forming foams.[1,2] Over the past several decades, long-chain PFAS, including perfluoroalkane sulfonic acids (PFSAs; $C_nF_{2n+1}SO_3H$, $n \geq$ six carbon atoms) and perfluoroalkyl carboxylic acids (PFCAs; $C_nF_{2n+1}COOH$, $n \geq$ seven carbon atoms), have attracted increasing scientific and regulatory attention owing to their environmental persistence, bioaccumulation, and potential toxicity in wildlife[3,4] and humans.[5,6] As such, various regulations have been implemented to restrict the manufacture and use of specific PFAS,[7–10] especially perfluorooctanesulfonate (PFOS)[11] and perfluorooctanoate (PFOA).[12] To replace these restricted PFAS, many manufacturers have introduced short-chain and partially fluorinated PFAS alternatives, characterized by the substitution of hydrogen or insertion of ether-oxygen in their backbones,[2,13,14] for example, per- and polyfluoroalkyl ether carboxylic acids and sulfonic acids (PFECAs and PFESAs).[15–17]

However, there is growing evidence that these novel compounds are also environmentally persistent and exhibit similar or even higher toxicity than PFOS and PFOA. In recent years, hexafluoropropylene oxide dimer acid (HFPO-DA; commercial name: GenX), hexafluoropropylene oxide trimer acid (HFPO-TA), and 6:2 chlorinated polyfluorinated ether sulfonate (6:2 Cl-PFESA; commercial name: F-53B) have been increasingly detected in the environment,[18,19] even in polar regions,[20,21] and in wildlife.[16,22] In addition, various emerging PFAS [i.e., HFPO-TA, perfluoro(3,5,7,9,11-pentaoxadodecanoic) acid (PFO5DoDA), and 6:2 Cl-PFESA] are reported to impact human liver function[23] and bioaccumulate in the estuarine food web.[24] Recent toxicological evidence also suggests that perfluoro-(3,5,7,9-tetraoxadecanoic) acid (PFO4DA) and PFO5DoDA may exert higher developmental toxicity to zebrafish than PFOA.[25] Of concern, however, the potentially harmful effects of many novel PFAS have not yet been verified in humans, including vulnerable populations, such as infants.

Given their immature metabolic and immune systems, infants are more susceptible than adults to risks associated with hazardous chemicals.[26] Human milk provides excellent nutrition and antibodies for the healthy development of infants[27,28] but is also considered a source of exposure to external pollutants,[29] potentially accounting for most PFAS intake in infants, including >94% and 83% of the exposure to PFOS and PFOA, respectively.[30] PFAS content in human milk has been reported globally, although previous studies have focused on legacy PFAS.[31–33] For example, studies have found that PFOS and PFOA are the most common PFAS found in samples of human milk from mothers in parts of the United States (median concentration = 30 ng/L of PFOS and

Address correspondence to Jiayin Dai, State Environmental Protection Key Laboratory of Environmental Health Impact Assessment of Emerging Contaminants, School of Environmental Science and Engineering, Shanghai Jiao Tong University, 800 Dongchuan Rd., Shanghai 200240, China. Telephone: 86-21-54741065. Email: daijy65@sjtu.edu.cn. And, Xiaoxia Bai, Women's Hospital, School of Medicine, Zhejiang University, Hangzhou, Zhejiang 310006, China. Telephone: 86-21-54741065. Email: baixiaoxia@zju.edu.cn

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14 ng/L of PFOA),[31] Czech Republic (20 ng/L and 23 ng/L),[34] Sweden (72 ng/L and 89 ng/L),[35] and Korea (47 ng/L and 39 ng/L).[29]

Nevertheless, few studies have reported on PFAS concentrations in human milk from Chinese mothers, except for small-scale local investigations in single cities, namely, Shanghai[35] and Hangzhou.[36] Animal model studies have shown that early postnatal PFAS exposure may lead to thyroid dysfunction,[37] decreased immunity,[38] and diseases in later life, such as obesity and diabetes.[39] Additional epidemiological evidence suggests that prenatal and postnatal PFAS exposure may increase the risk of poor growth in early childhood during the most sensitive stages of life.[40–42] Thus, it is essential to provide a national baseline of PFAS exposure in human milk in China and to assess the risks of emerging and legacy PFAS exposure in exclusively breastfed infants.

Studies on PFAS clearance and elimination half-life have been reported in highly exposed adult populations (e.g., fluorochemical production workers, airport employees) by biomonitoring serum samples across multiple longitudinal time points[43–45] or by analyzing paired serum and urine samples at a single time point,[46,47] with research on infants remaining scarce. In occupationally exposed workers, average renal clearance rates (CL_{renal}s) of PFOA, perfluorohexane sulfonate (PFHxS), and PFOS are 0.067, 0.023, and 0.010 mL/kg body weight (BW) per day, respectively.[46] Furthermore, PFOA (at 0.674 mL/kg BW per day) and PFOS (at 0.034 mL/kg BW per day) are eliminated more rapidly in the general population compared with occupationally exposed populations.[48] Thus, PFAS clearance estimates show considerable variation among different populations.

To evaluate the health risks of breast milk to infants, the accumulation and elimination behaviors of PFAS must be comprehensively considered given that infant organs are more sensitive to contaminants. To date, however, studies on renal clearance of PFAS in infants are limited. Despite the detection of PFAS in the serum, placenta, and breast milk of mothers, as well as in the cord blood of neonates, exploration of their associations in various paired biological matrices remains poor. One of the main reasons is that biological sample selection in birth cohorts is restricted, particularly body fluids from neonates (i.e., serum). Thus, cord blood is considered a good biological matrix reflecting internal PFAS exposure in infants at birth.[33] In addition, the difficulty in collecting urine samples from neonates has also hampered investigations on the elimination of hazardous chemicals from the body. To date, only a few studies have analyzed phthalate excretion in infants based on urine samples collected from disposable gel absorbent diapers.[49] To the best of our knowledge, however, studies have not reported on the occurrence and distribution of PFAS in multiple matched mother–infant samples. Therefore, a comprehensive assessment of postnatal PFAS exposure and accumulation via daily human milk intake and urinary excretion is lacking. At the same time, given the increase in emerging PFAS exposure, it is critical to understand the renal clearance of PFAS and predict their serum concentrations at key developmental stages in human infants.

In the present study, we aimed to *a*) determine the concentrations of 22 PFAS (including nine emerging PFAS) in human milk from lactating women obtained in the first 2 weeks postpartum in 21 cities in China, *b*) estimate the daily intake of PFAS and net accumulation exposure to PFAS in nursing infants, and *c*) calculate the renal clearance of PFAS in three paired biological samples (cord blood, human milk, and urine) and predict PFAS serum concentrations using a single compartment and first-order pharmacokinetic model. This study should not only provide health protection thresholds for breastfeeding infants but also provide information on the elimination and half-life of emerging PFAS, which can be used to compare the potential health risks of emerging and legacy PFAS.

Methods

Study Population and Sample Collection

Expectant mothers were recruited prior to delivery in the local hospital. The mothers were informed regarding the breast milk study plan and purpose in the hospital before delivery. Those willing to participate in the study provided fully informed consent for our use of their samples and study aim. The criteria for recruitment were minimal being able to *a*) produce breast milk after delivery (including both vaginal and cesarean section) and *b*) complete a basic questionnaire. Mothers suffering miscarriage, stillbirth, or delivery complications/injury, such as amniotic fluid embolism and placental abruption, were excluded from recruitment. From October 2020 to June 2021, a total of 1,151 postpartum milk samples were provided by participants from 21 cities in China (Table S1 and Figure S1). Breast milk samples were collected from two hospitals in Shanghai and one hospital in each of the other study cities. Lactating women within the 2-week postpartum period were instructed to express breast milk manually into a centrifuge tube. During the collection period, breast milk samples were stored at −80°C at the local hospital. Once the collection process was completed, the samples were transported on dry ice to our laboratory at Shanghai Jiao Tong University for PFAS measurement. Of the 1,151 participants, those from Huantai County ($n = 15$) and Hangzhou City ($n = 65$) provided additional paired cord blood and infant urine samples. Cord blood samples were collected during delivery. Disposable diapers containing urine were collected in the hospital obstetrics department during the first postnatal week and sealed in polypropylene bags. All samples were stored at −20°C within 24 h of collection and transferred to −80°C until further analysis.

All mothers provided fully informed consent for our usage of their samples and completed a questionnaire with the assistance of well-trained nurses. The questionnaire included information on maternal age (in years; continuous), postnatal weight (in kilograms; continuous), height (in meters; continuous), education (high school, bachelor's degree, advanced degree; categorical), annual family income [Chinese Yuan Renminbi (CNY); low: <50,000, middle: 50,000–100,000, upper middle: >100,000; categorical)], hyperglycemia (yes or no; categorical), hypertension (yes or no; categorical), type of delivery (cesarean section or vaginal delivery; categorical), gravidity (primiparous or multiparous; categorical), infant sex (boy or girl; categorical), birth weight (in grams; continuous), and birth length (in centimeters; continuous). All research protocols and ethics applications were approved by the participating hospitals. The ethics committee of Shanghai Jiao Tong University provided approval for this study.

Standards and Reagents

A total of 22 target PFAS, including 9 emerging PFAS [i.e., perfluoro-2-methoxyacetic acid (PFMOAA), perfluoro(3,5-dioxahexanoic) acid (PFO2HxA), perfluoro(3,5,7-trioxaoctanoic) acid (PFO3OA), PFO4DA, PFO5DoDA, HFPO-DA, HFPO-TA, and chlorinated polyfluorinated ether sulfonates (6:2 and 8:2 Cl-PFESA)] and 13 legacy PFAS (C4–C12 PFCAs and C4, C6–C8 PFSAs), were analyzed in this study. Except for PFMOAA, PFO2HxA, PFO3OA, PFO4DA, PFO5DoDA, and HFPO-TA, all native and mass-labeled internal standards (purity >99%; listed in Table S3) were purchased from Wellington Laboratories. The remaining native standards (purity >98%) were synthesized using previously reported methods.[23]

Ammonium acetate (≥99.9%) was purchased from Sigma. Methanol [liquid chromatography–mass spectrometry (LC-MS) grade], acetonitrile [ACN; high-pressure LC (HPLC) grade], water (LC-MS grade), formic acid, acetic acid, and ammonium hydroxide were obtained from Fisher Scientific. *N*-propylethylenediamine (PSA) adsorbent was purchased from Agilent Technologies. Calcium chloride ($CaCl_2$; ≥96%) was purchased from Shanghai Macklin Biochemical Co., Ltd. Oasis weak anion exchange (WAX) cartridges (6 cc/150 mg) were purchased from the Waters Corporation.

PFAS Measurement

The target compounds were measured in human milk, cord blood, and urine based on previous extraction methods, with minor modifications.[23,31,50] In brief, 2 mL of breast milk was extracted using 2 mL of acidified ACN (containing 1% formic acid), with the process repeated twice. The supernatants were combined and concentrated to near dryness (<0.5 mL) under nitrogen (N_2). The remaining 0.5 mL of solution was diluted with 10 mL of 1% formic acid water and further purified using a solid phase extraction (SPE) cartridge (Oasis WAX 6 cc/150 mg; Waters). The cartridge was preconditioned with 8 mL of 0.5% ammonium hydroxide in methanol, 4 mL of methanol, and 4 mL of ultrapure water. After loading the samples, the cartridges were washed with 4 mL of buffer solution (25 mM acetic acid/ammonium acetate, pH = 4) and 4 mL of methanol. The target compounds were eluted using 4 mL of 0.5% ammonium hydroxide in methanol. The eluent was evaporated under N_2 and reconstituted with 200 μL of methanol/water (vol/vol = 50:50) for instrumental analysis.

Cord blood was extracted using ACN for protein precipitation.[23] Briefly, 200 μL of blood was spiked with 5 ng of internal standard and extracted with 1 mL of ACN. The mixture was vortex-mixed and sonicated for 10 min. The supernatant was transferred into a new 15-mL tube after vortexing for 10 min at 900 rpm and centrifuged for 15 min at 148,000 rpm and 4°C. Another 1 mL of ACN was added to the remaining tube, and the extraction procedure was repeated as described above. The extract solution (2 mL) was evaporated to dryness under N_2 at 40°C and reconstituted with 200 μL of methanol/water (vol/vol = 50:50).

Urine samples were extracted from the gel diaper by mixing the gel absorbent with $CaCl_2$, as per previously reported methods,[50] with a customized glass cup used to separate urine from the dehydrated gel absorbent. In detail, the wet diaper was cut open after thawing at room temperature. The gel absorbent particles were carefully transferred from the diaper to a glass cup with a filter at the bottom. An amount of $CaCl_2$ powder was added to the glass cup and mixed thoroughly with the gel absorbent (ratio of $CaCl_2$ to gel absorbent, 1:50 wt/wt). The mixture was then incubated for 15 min at room temperature to release the urine from the gel absorbent. The tube was centrifuged at 3,000 rpm for 3 min and a "clear" urine sample was collected for further preparation. The urine sample (10 mL) was concentrated using an Oasis WAX cartridge following the same procedure as for breast milk. Additional cleanup was performed by adding 50 mg of PSA absorbents before evaporation. The supernatant was evaporated under N_2 and reconstituted with 200 μL of methanol/water (vol/vol = 50:50) for instrumental analysis. Considering further computational convenience for PFAS elimination and half-life estimation, as well as limitations of urine sampling, PFAS concentrations were not adjusted for creatinine concentration or specific gravity.

The extractants were finally injected into an ultra-high-performance liquid chromatograph (ExionLC AD; Applied Biosystems/SCIEX Inc.) coupled to an electrospray ionization tandem mass spectrometer (QTRAP 6500 plus; AB SCIEX) for quantification of PFAS. Owing to the fragmentation of PFECAs in the ion source, another quantification method was used exclusively for PFECAs. Details on chromatographic columns, instrumental parameters, and tandem mass spectrometry parameters are presented in Tables S2 and S3.

Quality Assurance and Quality Control

Potential laboratory background, field blank, and solvent blank contaminations were checked before processing formal samples. One procedural blank (Milli-Q water) and one quality control sample (Standard Reference Material, SRM 1957, used for cord blood and mixed and spiked human milk and urine samples used for human milk and infant urine, respectively) were used every 20 samples to monitor background contamination and ensure the accuracy of data from each batch. PFAS were not detected in the procedural blanks ($n = 70$), except for HFPO-DA and PFBA (means = 2.23 ng/L and 1.68 ng/L, respectively, in the blanks). Thus, the levels of HFPO-DA and PFBA were blank-corrected by subtracting the average procedural blank concentrations from the sample concentrations. An 11-point internal calibration curve (0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2, 5, 10, and 20 ng/mL) was prepared for all analytes in matrix-free solvent, which exhibited excellent linearity ($R^2 > 0.99$), except for PFMOAA (1–20 ng/mL) in the urine analysis matrix calibration curve.

Instrumental drift was monitored by injecting a calibration standard every 10 sample injections, and a new calibration curve was constructed if a ±20% deviation from its initial value was observed. The limits of detection (LODs) were set as the PFAS concentration resulting in a matrix-specific signal-to-noise (S/N) ratio of 3. The method limits of quantification (LOQs) were established based on the concentration resulting in a matrix-specific S/N ratio of 10 or the lowest concentration in the calibration curve with measured concentrations within ±20% of its theoretical value, followed by adjustment with the concentration or dilution factor in extraction. Details on LODs and LOQs in different matrices are provided in Table S4. Matrix-spiked recoveries ($n = 4$) were assessed by spiking 0.2 ng of mixed native standards (final concentration 1.0 ng/mL) into a blank matrix (pooled human milk, infant cord blood, and infant urine), followed by the same analytical methods above. The recoveries of PFAS in human milk, cord blood, and urine samples ranged from 78% to 114% (Table S5).

Data Analysis

Descriptive statistics were assessed for participant demographics, neonatal birth information, and PFAS concentrations in three biomatrices (human milk, cord blood, and urine). In calculations of average proportions among all PFAS in the biomatrices, PFAS concentrations below LOQ were set to zero, and the proportions were calculated via two steps: *a*) contribution percentage of PFAS in each sample, and *b*) average percentage of individual PFAS across all human milk samples.

Considering the nonnormality of data, the Mann–Whitney *U*-test and Kruskal–Wallis test were used to compare differences in PFAS concentrations in human milk samples between demographic variables. Maternal age (continuous variable) was divided to four groups (<25, 25–29, 30–34, ≥35; categorized variable). Maternal weight (continuous variable) and height (continuous variable) were calculated to postnatal body mass index (BMI; categorized variable) based on Chinese BMI standards (underweight: <18.5 kg/m^2, normal: 18.5–23.9 kg/m^2, overweight: ≥24 kg/m^2). For missing data on demographic characteristics, PFAS concentrations in these samples were excluded, and comparisons of PFAS concentrations were analyzed in other subgroups. Concentrations below the LOQs were substituted by the LOQ divided by 2 for analytes with detection rates of >50% in human milk samples for analysis of differences. Other PFAS with low detection rates in human milk [i.e., HFPO-DA,

HFPO-TA, PFO2HxA, PFO3OA, PFO4DA, 8:2 Cl-PFESA, perfluoropentanoate (PFPeA), perfluorohexanoate (PFHxA), perfluorobutane sulfonate (PFBS), and perfluoroheptane sulfonate (PFHpS)] were not further explored.

Hierarchical cluster analysis was used to assess distribution similarities or differences in PFAS concentrations between various sampling cities. The complete-linkage approach was used for clustering to avoid the generation of links in single linkages.[51] The distance between clusters was defined as the longest distance between clusters. Hierarchical cluster analysis was conducted using TBtools (open source software; https://github.com/CJ-Chen/TBtools/releases). Based on the clustering results, the various sampling cities were divided into different regions. Significant differences in individual PFAS within clusters were verified using the Kruskal–Wallis test. Statistical analyses were performed using SPSS (version 24.0; SPSS Inc.), and the significance threshold was set to $p < 0.05$.

The estimated daily intake (EDI; in nanograms per kilogram BW per day) of each PFAS was calculated for breastfed infants using the following equation:

$$EDIs = \frac{C_{BM} \times V_{BM}}{BW},$$

where C_{BM} represents PFAS concentration measured in breast milk (in nanograms per liter); V_{BM} is the average daily consumption volume of breast milk (510 mL/d for infants <1 month of age)[52]; and BW is the body weight of the infant at birth (in grams). BW was assumed as 3,400 g for infants missing birth weight data, based on average food ingestion rate (FIR; 150 mL/kg BW per day) and V_{BM} (510 mL/d) published in the *Exposure Factors Handbook*.[52] In total, 78 infants were missing birth weight data, including 41 in Dalian, 3 in Huantai, 2 in Shouguang, 6 in Shanghai, and 26 in Guangzhou. Sensitivity analysis was conducted by repeating EDI calculations after excluding those with missing birth weight. Independent-samples *t*-tests were used to assess differences in PFAS EDIs based on infant sex.

The daily absolute intake/excretion mass of PFAS (*AIM*/*AEM*; in nanograms per day) and initial body burden of PFAS exposure (E_{IB}; in nanograms) for infants at birth were calculated using the following equations:

$$AIM = \frac{C_{BM} \times FIR \times BW}{1{,}000 \times 1{,}000},$$

$$AEM = \frac{C_{UR} \times UER \times BW}{1{,}000 \times 1{,}000},$$

$$E_{IB} = \frac{C_{CB} \times V_B \times BW}{1{,}000 \times 1{,}000},$$

where C_{UR} is the concentration of PFAS in urine (in nanograms per liter); UER (urine excretion rate) is the average daily excretion via urine, set to 48 mL/kg BW per day[53]; C_{CB} is the concentration of PFAS in cord blood (in nanograms per liter); and V_B is the blood volume in the body, estimated as 85 mL/kg BW for infants <3 months of age.[54] The two factors (1,000) in the denominator of the equations were used for unit conversion, representing the conversion of milliliters per kilogram BW per day (or milliliters per kilogram BW) to liters per kilogram BW per day (or liters per kilogram BW) for intake/excretion rate (or blood volume) and grams to kilograms for BW. Here, we assumed that PFAS levels in cord blood represent internal exposure of infants at birth, given that cord blood is present on the fetal side of the vascular organ.[55,56] The correlation analysis of PFAS concentrations was explored between breast milk and cord blood samples and between breast milk and urine samples using regression analyses.

We calculated the daily CL_{renal} (in milliliters per kilogram BW per day) of individual PFAS on the basis of paired blood and urine concentrations, and we estimated the elimination half-lives ($T_{1/2}$, in days) using the following equations.[48] As in Zhang et al. (2013),[48] we assumed renal clearance to be the major pathway for PFAS elimination to simplify half-life estimations.

$$CL_{renal} = \frac{C_{UR} \times UER}{2 \times C_{CB}},$$

$$T_{1/2} = \frac{\ln(2) \times V_d}{CL_{renal}},$$

where the factor of 2 in the denominator of the equation represents the PFAS concentrations in serum-to-whole blood ratio[57,58] and V_d is the apparent volume of distribution (in liters per kilogram). The V_d values of PFOS and PFOS were set to 230 and 170 mL/kg, respectively.[59] Comparable V_d values of PFOA have been reported in different mammals [mean ± standard deviation (SD), 191 ± 67 mL/kg].[60] Owing to the lack of V_d values for PFAS in humans, we tentatively used the toxicokinetic parameters of rats and mice for the calculation of half-life. We set 130, 199, 265, and 394 mL/kg as the V_d values for PFBA, PFHpA, PFNA, and PFDA, respectively.[61,62] The V_d values of PFO3OA, PFO4DA, and PFO5DoDA were set to 103, 154, and 296 mL/kg based on toxicokinetic data of mice.[63] The V_d values of other PFECAs and PFCAs were estimated based on the V_d of substances with similar structures, such as 103 mL/kg for PFMOAA, PFO2HxA, PFO3OA, and HFPO-DA; 296 mL/kg for HFPO-TA; and 130 mL/kg for PFPeA. Except for the V_d value of PFBS (277 mL/kg) estimated in CD-1 mice,[64] we assumed a V_d value of 230 mL/kg for other PFSAs to maintain consistency with other studies.[46] We further compared the estimated CL_{renal}s in the present work to published data in prior studies. Available data on renal PFAS clearance in humans were identified by searching keywords "PFAS" and "RENAL CLEARANCE" in PubMed and Web of Science for publications appearing between 1 January 2000 and 31 December 2021.

Given that the one-compartment model has been previously adopted to estimate time-dependent serum concentrations and PFAS compounds are not highly lipophilic,[59,65] a first-order model based on the one-compartment model was employed to address temporal trends in PFAS in infants in days after birth (C_t) via breast milk intake[66]:

$$\frac{dC_t}{dt} = -k \times C_t + \frac{EDIs}{V_d},$$

where k is the elimination rate calculated using the half-life with the following equation:

$$k = \frac{ln(2)}{T_{1/2}}.$$

The solution for C_t can be expressed as follows:

$$C_t = \frac{EDIs}{k \times V_d}(1 - e^{-k \times t}) + C_0 \times e^{-k \times t},$$

where C_t is the serum concentration in infants after birthday t, and C_0 is the initial serum concentration in infants at birth. As above, PFAS levels in cord blood were regarded as the initial blood concentration in infants. Thus, PFAS concentrations in whole blood were multiplied by a factor of 2 to convert to a

Table 1. Summary of demographic characteristics of study participants ($n = 1{,}151$).

Parameter	n	Percentage (%)
Maternal age (y)		
<25	107	9.5
25–29	414	36.7
30–34	398	35.3
≥35	209	18.5
Missing	23	—
Education		
High school	657	60.2
Bachelor's degree	371	34.0
Advanced degree	63	5.8
Missing	60	—
Annual family income (CNY)		
Low (<50,000)	74	8.4
Middle (50,000–100,000)	336	38.1
Upper middle (>100,000)	471	53.5
Missing	270	—
Type of delivery		
Cesarean section	644	56.3
Vaginal delivery	500	43.7
Missing	7	—
Gravidity		
Primiparous	636	55.9
Multipara	501	44.1
Missing	14	—
Postnatal BMI (kg/m^2)		
Underweight (<18.5)	26	2.4
Normal (18.5–24.0)	354	32.5
Overweight (>24.0)	710	65.1
Missing	61	—
Hyperglycemia		
Yes	180	15.8
No	959	84.2
Missing	12	—
Hypertension		
Yes	177	15.9
No	933	84.1
Missing	41	—
Infant sex		
Girl	493	45.0
Boy	603	55.0
Missing	55	—
Infant weight (g)		
<2,500	106	9.9
2,500–4,000	874	81.5
>4,000	93	8.7
Missing	78	—

Note: The number of missing was removed from the calculation of percentages. —, not applicable; BMI, body mass index; CNY, Chinese Yuan Renminbi.

serum concentration before substituting the equation.[54] A factor of 2 was used for the ratio of PFAS concentrations in serum to whole blood based on previous studies of the general adult population.[57,58] The prediction models for PFAS concentrations in serum of infants <1 year of age were generated using R (version 4.0.2; R Development Core Team).

Results

Population Characteristics

A summary of the demographic characteristics and basic information of the 1,151 mothers and their infants is provided in Table 1. In total, 98% of participants provided their maternal age, which ranged from 16 to 47 years of age (mean ± SD, 30.3 ± 4.7 y). Nearly half of the participants had a bachelor's degree or above, with an upper middle-class income (>1 00,000 CNY). More than 50% of mothers were primiparous and chose to have a cesarean section. The average postpartum BMI value was 25.6 kg/m^2, and 65% of participants were overweight. Most mothers (>80%) did not suffer from hyperglycemia or hypertension during pregnancy. Of the neonates with known sex, males accounted for 55% and females accounted for 45%. The mean ± SD of birth weight was 3,277 ± 708 g, with 82% of infants within the normal weight range.

PFAS Concentrations in Breast Milk

The detection rates and descriptive statistics of PFAS concentrations in human milk are summarized in Table 2. Among the 22 PFAS detected in milk, 3 emerging PFAS (PFMOAA, PFO5DoDA, and 6:2 Cl-PFESA) and 9 legacy PFAS were detected in more than half of the samples (Figure 1, Table 2, and Excel Table S1). The other PFAS analytes were detected in 8.5%–42.7% of samples and hence were not further analyzed. Among all PFAS analytes, PFOA was dominant (median = 336 ng/L), accounting for 63% of total PFAS (on average), followed by PFOS (49.7 ng/L, 13%) and 6:2 Cl-PFESA (13.6 ng/L, 4.5%). To the best of our knowledge, 2 emerging PFECAs (PFMOAA and PFO5DoDA) were found in human milk for the first time, with median concentrations comparable to those of PFBA and PFHpA. We examined differences in PFAS concentrations and distribution among demographic subgroups (Table S6). The concentrations of several long-chain PFAS, such as PFO5DoDA, 6:2 Cl-PFESA, and PFOS, were significantly higher in older and upper-middle income groups. Higher PFAS levels, except for PFOA, were also observed in human milk from mothers with a bachelor's degree. In contrast, the highest PFOA concentrations were found in mothers with a high school diploma and low annual family income. Notably, milk concentrations of four long-chain PFAS (PFO5DoDA, 6:2 Cl-PFESA, PFOA, and PFOS) were significantly higher in mothers with gestational hypertension. The median concentrations of PFO5DoDA, 6:2 Cl-PFESA, and PFOA were about 1.4 times higher in postpartum overweight females than in the normal-weight group.

PFAS Profiles and Clustering in Various Cities

Figure 2A shows the similarities in PFAS composition in human milk samples across all sampled cities (corresponding numeric results are provided in the Excel Table S2). Median concentrations (in nanograms per liter) of PFAS (detection rate >50%) in breast milk obtained in individual cities are summarized in Table S7. Although levels varied in the different cities, PFOA accounted for the largest proportion of total PFAS in all areas, ranging from 37% (Taizhou) to 87% (Guangzhou). PFOS was the second-most common compound, ranging from 2.2% (Huantai County) to 32% (Wuhan), although a higher proportion of PFECAs was detected in several specific areas (e.g., Sanming, Huantai, Shouguang). The percentage contributions of several long-chain PFCAs (PFDA, PFUnDA, and PFDoDA) and PFHxS were higher in eastern China, including Shanghai, Nanjing, Hangzhou, Huzhou, Taizhou, Jiaxing, Hefei, Anqing, Wenzhou, and Ningbo, than in other cities, whereas the proportions of PFBA and PFHpA were higher in Xiamen, Sanming, and Wuhan. The remaining PFAS (PFPeA, PFHxA, PFBS, and PFHpS) were found at negligible percentages in the studied locations. The cities were divided into four clusters based on PFAS composition using complete-linkage cluster analysis. As shown in Figure 2B, the median concentrations of PFO5DoDA in human milk differed significantly in the four clusters. PFMOAA levels were highest in Region 1, whereas PFOA levels were significantly higher in Regions 1 and 3. Highly elevated concentrations of 6:2 Cl-PFESA, PFOS, and PFNA were observed in Region 4. The raw data are provided in the Excel Table S1 and p-values from different comparisons of the four regions are reported in Table S8.

Table 2. Concentrations of PFAS (ng/L) in human milk ($n = 1{,}151$).

Analyte	n	DR (%)	GM	SD	Min	Percentile					Max	Proportion (%)[a]
						5th	25th	50th	75th	95th		
Emerging PFAS												
HFPO-DA	98	8.5	3.056	10.09	<LOQ	<LOQ	<LOQ	<LOQ	<LOQ	5.527	199.1	0.2
HFPO-TA	212	18.4	1.783	22.52	<LOQ	<LOQ	<LOQ	<LOQ	<LOQ	13.41	524.3	0.2
PFMOAA	1,099	95.5	4.816	75.34	<LOQ	1.042	1.946	3.416	7.411	107.0	1,086	2.4
PFO2HxA	208	18.1	0.733	7.887	<LOQ	<LOQ	<LOQ	<LOQ	<LOQ	8.604	125.5	0.2
PFO3OA	492	42.7	1.275	34.27	<LOQ	<LOQ	<LOQ	<LOQ	2.600	21.24	744.4	0.8
PFO4DA	343	29.8	1.763	11.03	<LOQ	<LOQ	<LOQ	<LOQ	2.659	15.82	144.9	0.4
PFO5DoDA	820	71.2	2.380	9.002	<LOQ	<LOQ	<LOQ	2.633	5.731	17.78	117.5	0.7
6:2 Cl-PFESA	1,147	99.7	15.24	75.70	<LOQ	2.886	6.923	13.59	32.02	104.3	1,183	4.5
8:2 Cl-PFESA	266	23.1	0.696	5.224	<LOQ	<LOQ	<LOQ	<LOQ	<LOQ	3.561	163.3	0.1
Legacy PFAS												
PFBA	959	83.3	4.872	15.62	<LOQ	<LOQ	2.716	4.549	7.784	24.93	258.5	1.6
PFPeA	168	14.6	0.640	2.395	<LOQ	<LOQ	<LOQ	<LOQ	<LOQ	3.609	38.52	0.1
PFHxA	373	32.4	0.867	9.713	<LOQ	<LOQ	<LOQ	<LOQ	1.317	6.789	238.0	0.4
PFHpA	925	80.4	2.095	4.647	<LOQ	<LOQ	1.173	2.093	3.862	11.81	58.29	0.5
PFOA	1,151	100	325.7	1,129	12.04	58.07	144.9	335.9	668.9	1,933	2,059	63.0
PFNA	1,151	100	12.71	21.42	1.161	3.799	7.396	12.28	20.51	50.37	307.0	3.1
PFDA	1,142	99.2	9.343	38.88	<LOQ	1.893	4.379	8.721	17.92	59.24	798.3	2.6
PFUnDA	1,144	99.4	9.244	22.09	<LOQ	2.459	5.025	8.598	15.81	43.87	438.2	2.4
PFDoDA	744	64.6	1.523	6.510	<LOQ	<LOQ	<LOQ	1.471	2.949	10.25	135.2	0.4
PFBS	291	25.3	0.696	1.492	<LOQ	<LOQ	<LOQ	<LOQ	<LOQ	2.788	30.42	0.1
PFHxS	1,092	94.9	7.619	38.23	<LOQ	<LOQ	3.514	7.286	16.86	54.93	533.8	2.6
PFHpS	324	28.1	0.712	1.437	<LOQ	<LOQ	<LOQ	<LOQ	1.063	2.531	29.29	0.1
PFOS	1,151	100	52.83	154.6	2.621	13.81	29.38	49.72	87.79	257.4	2,503	13.7

Note: Cl-PFESA, chlorinated polyfluorinated ether sulfonates; DR, detection rate; GM, geometric mean; HFPO-DA, hexafluoropropylene oxide dimer acid; HFPO-TA, perfluoro-2,5-dimethyl-3,6-dioxanonanoic acid; LOQ, limit of quantitation; Max, maximum; Min, minimum; PFAS, per- and polyfluoroalkyl substances; PFBA, perfluorobutanoate; PFBS, perfluorobutane sulfonate; PFDA, perfluorodecanoate; PFDoDA, perfluorododecanoate; PFHpA, perfluoroheptanoate; PFHpS, perfluoroheptane sulfonate; PFHxA, perfluorohexanoate; PFHxS, perfluorohexane sulfonate; PFMOAA, perfluoro-2-methoxyacetic acid; PFNA, perfluorononanoate; PFO2HxA, perfluoro(3,5-dioxahexanoic) acid; PFO3OA, perfluoro(3,5,7-trioxaoctanoic) acid; PFO4DA, perfluoro(3,5,7,9-tetraoxadecanoic) acid; PFO5DoDA, perfluoro(3,5,7,9,11-pentaoxadodecanoic) acid; PFOA, perfluorooctanoate; PFOS, perfluorooctane sulfonate; PFPeA, perfluoropentanoate; PFUnDA, perfluoroundecanoate; SD, standard deviation.
[a]Average proportion among all PFAS in human milk samples. Proportions were calculated by two steps: namely, determined contribution percentage of PFAS in each sample, then averaged percentages of individual PFAS across all human milk samples. PFAS concentrations below LOQ were set to zero.

Daily Intake of PFAS via Breast Milk

The PFOA and PFOS EDIs in individual cities are shown in Figure 3 and Figures S2, respectively (corresponding numeric data are provided in the Excel Tables S3 and S4). Descriptive statistics of EDIs for PFOA and PFOS and their differences by neonatal sex are summarized in Table S9. We found that the EDIs of PFOA and PFOS exceeded the reference dose (RfD) of 20 ng/kg BW per day recommended by the U.S. Environmental Protection Agency (EPA) in 78% and 17% of breastfed infant samples in all cities, respectively. Furthermore, more than half of the PFOA EDIs were above the threshold in 14 cities, including Huantai, Shouguang, Guangzhou, and Quzhou, in which the EDIs exceeded the recommended threshold (20 ng/kg BW per day) in all tested infants.[67] In comparison, in half the cities, only 20% of the PFOS EDIs exceeded the RfD (20 ng/kg BW per

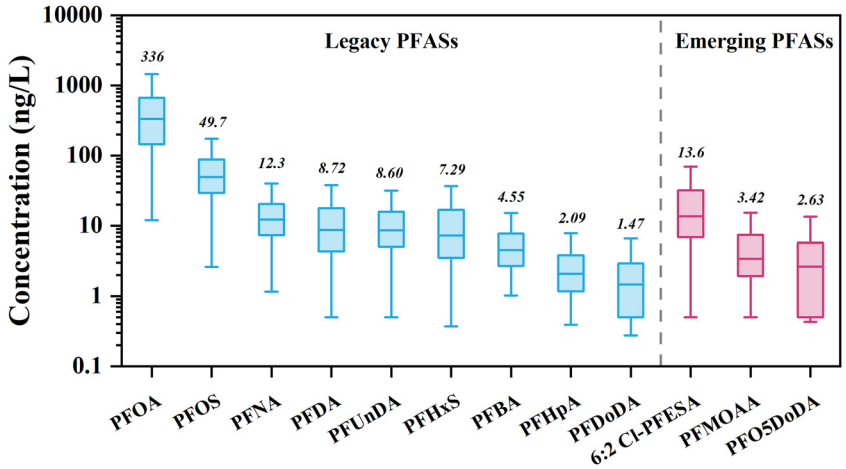


Figure 1. Concentrations (ng/L) of PFAS detected in >50% of human milk samples ($n = 1{,}151$). Boxes display 25th, 50th, and 75th percentiles for PFAS concentrations, and whiskers represent 10th and 90th percentiles. Values above box represent median concentration. Corresponding raw data are provided in Excel Table S1. Boxes are ranked from highest to lowest median level for each PFAS. Blue boxes (left of dashed line) represent legacy PFAS, pink boxes (right of dashed line) represent emerging PFAS. Note: Cl-PFESA, chlorinated polyfluorinated ether sulfonates; PFAS, per- and polyfluoroalkyl substances; PFBA, perfluorobutanoate; PFDA, perfluorodecanoate; PFDoDA, perfluorododecanoate; PFHpA, perfluoroheptanoate; PFHxS, perfluorohexane sulfonate; PFMOAA, perfluoro-2-methoxyacetic acid; PFNA, perfluorononanoate; PFO5DoDA, perfluoro(3,5,7,9,11-pentaoxadodecanoic) acid; PFOA, perfluorooctanoate; PFOS, perfluorooctane sulfonate; PFUnDA, perfluoroundecanoate.

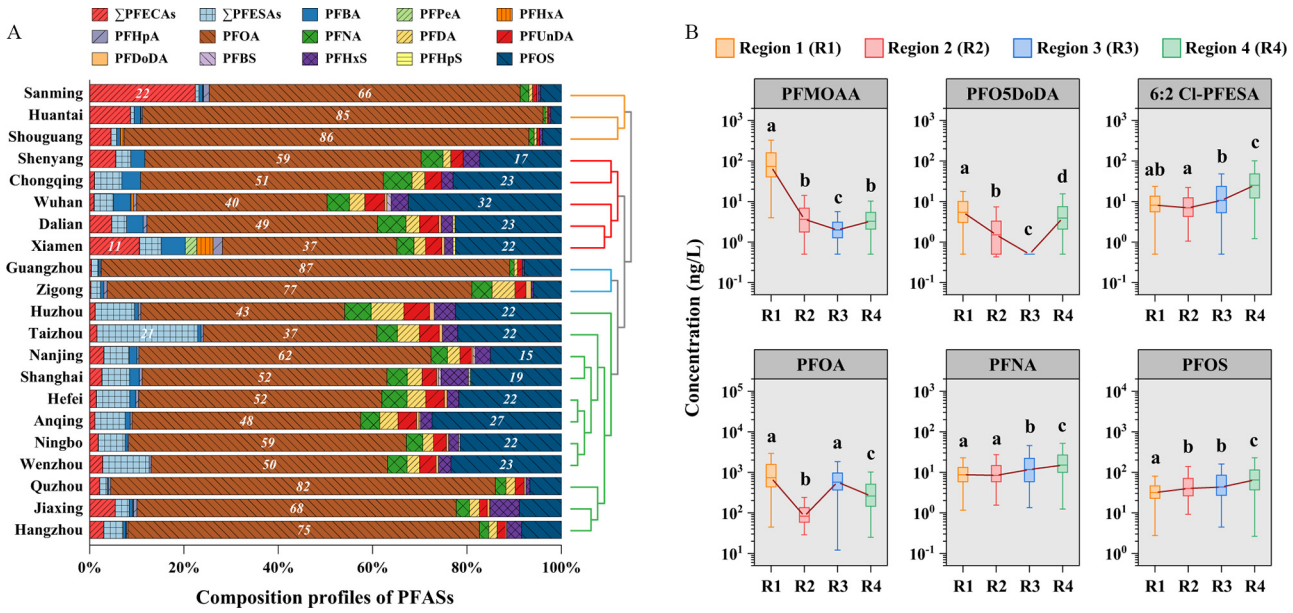


Figure 2. Composition profiles and concentrations of PFAS in human milk samples from different cities and regions. PFAS median concentrations were used to calculate composition profiles in each city. (A) Hierarchical cluster analysis was performed on profiles of 21 cities using complete-linkage cluster analysis. Cities were divided into four clusters. (B) Box and whisker plots for PFMOAA, PFO5DoDA, 6:2 Cl-PFESA, PFOA, PFNA, and PFOS concentrations in regions based on clustering results. Region 1 (R1, orange boxes, $n = 169$) included Sanming, Huantai, and Shouguang. Region 2 (R2, red boxes, $n = 172$) included Shenyang, Chongqing, Wuhan, Dalian, and Xiamen. Region 3 (R3, blue boxes, $n = 259$) included Guangzhou and Zigong. Region 4 (R4, green boxes, $n = 551$) included Huzhou, Taizhou, Nanjing, Shanghai, Hefei, Anqing, Ningbo, Wenzhou, Quzhou, Jiaxing, and Hangzhou. Different letters represent significant differences between groups at $p < 0.05$ by Kruskal–Wallis test. All specific p-values from comparisons are reported in Table S8. Corresponding raw data are provided in Excel Tables S1 and S2. ∑PFECAs represent sum of HFPO-DA, HFPO-TA, PFMOAA, PFO2HxA, PFO3OA, PFO4DA, and PFO5DoDA; ∑PFESAs represent sum of 6:2 Cl-PFESA and 8:2 Cl-PFESA. Note: Cl-PFESA, chlorinated polyfluorinated ether sulfonates; HFPO-DA, hexafluoropropylene oxide dimer acid; HFPO-TA, perfluoro-2,5-dimethyl-3,6-dioxanonanoic acid; PFAS, per- and polyfluoroalkyl substances; PFBA, perfluorobutanoate; PFBS, perfluorobutane sulfonate; PFDA, perfluorodecanoate; PFDoDA, perfluorododecanoate; PFECAs, per- and polyfluoroalkyl ether carboxylic acids; PFESAs, per- and polyfluoroalkyl ether sulfonic acids; PFHpA, perfluoroheptanoate; PFHpS, perfluoroheptane sulfonate; PFHxA, perfluorohexanoate; PFHxS, perfluorohexane sulfonate; PFMOAA, perfluoro-2-methoxyacetic acid; PFNA, perfluorononanoate; PFO2HxA, perfluoro(3,5-dioxahexanoic) acid; PFO3OA, perfluoro(3,5,7-trioxaoctanoic) acid; PFO4DA, perfluoro(3,5,7,9-tetraoxadecanoic) acid; PFO5DoDA, perfluoro(3,5,7,9,11-pentaoxadodecanoic) acid; PFOA, perfluorooctanoate; PFOS, perfluorooctane sulfonate; PFPeA, perfluoropentanoate; PFUnDA, perfluoroundecanoate; R, region.

day),[68] whereas in other cities, levels were below the reference threshold in all tested samples. Of concern, when compared with stricter reference values, average PFOA intake via human breast milk was more than two orders of magnitude higher than the tolerable daily intake (0.857 ng/kg BW per day) established by the EFSA (European Food Safety Authority).[69] Of note, all PFOA EDIs exceeded this reference value in all cities. In addition, almost all PFOS EDIs (>90% samples) were higher than the EFSA reference dose (1.857 ng/kg BW per day).[69] Furthermore, 100% of PFOS EDIs exceeded 1.857 ng/kg BW per day in one-third of the cities (including Dalian, Xiamen, Huzhou, Nanjing, Anqing, Quzhou, and Jiaxing).

PFAS Mass Balance via Human Milk and Urine

The distribution of PFAS in paired cord blood and neonate urine samples ($n = 80$) is shown in Tables S10 and S11 and Excel Table S5. Except for HFPO-DA, PFO2HxA, 8:2 Cl-PFESA, and PFBA, the detection rates of all target analytes were >75% in cord blood, and the median concentrations ranged from 15.65 ng/L (in PFPeA) to 5,185 ng/L (in PFOA). The dominant PFAS in cord blood were the same as in human milk (PFOA > PFOS > 6:2 Cl-PFESA). The PFAS levels in neonatal urine were much lower than the levels in cord blood. Several long-chain PFAS, such as PFO4DA, PFO5DoDA, 8:2 Cl-PFESA, PFUnDA, PFDoDA, PFHpS, and PFOS, were not detected in any urine sample.

In contrast, short-chain PFBA and PFMOAA were detected in 100% of urine samples at concentrations 3–9 times higher than that of PFOA. The different PFAS proportions in the three biomatrices and the PFAS mass balance estimates are shown in Figure 4 and Figures S3–S4, respectively (corresponding numeric data are provided in the Excel Tables S6 and S7). Similar proportions of PFAS were detected in human milk and cord blood, but not in urine. For example, PFOA, PFOS, and 6:2 Cl-PFESA accounted for >70% of total PFAS in human milk and cord blood, whereas perfluorobutanoate (PFBA) and PFMOAA accounted for >70% of total PFAS in urine.

In the mass balance estimation model, the absolute daily mass of most PFAS detected in human milk and urine, except for PFMOAA, PFBA, and PFBS, showed a significant net accumulation in neonates (Figure 4 and Figure S3). Correlation analysis of the dominant legacy PFAS (i.e., PFOA, PFNA, and PFOS) and emerging PFAS (i.e., PFMOAA, PFO5DoDA, and 6:2 Cl-PFESA) in different matrices revealed significant positive correlations ($0.24 \le r^2 \le 0.78$) between individual PFAS levels in cord blood and human milk (Figure S5 and Table S12). Given that PFO5DoDA and PFOS were not detected in urine, we were unable to explore their associations between human milk and urine. The four other PFAS ($0.01 \le r^2 \le 0.29$) showed no significant correlations between human milk and urine.

Estimation of Renal Clearance and Half-Life

Table 3 shows the daily CL_{renal} (in milliliters per kilogram BW per day) of individual PFAS based on paired cord blood and urine concentrations. Among the PFCAs with C4 to C10 chain lengths, the median CL_{renal} decreased from 31.80 to 0.022 mL/kg BW per day as the perfluorinated carbon chain length increased. PFMOAA,

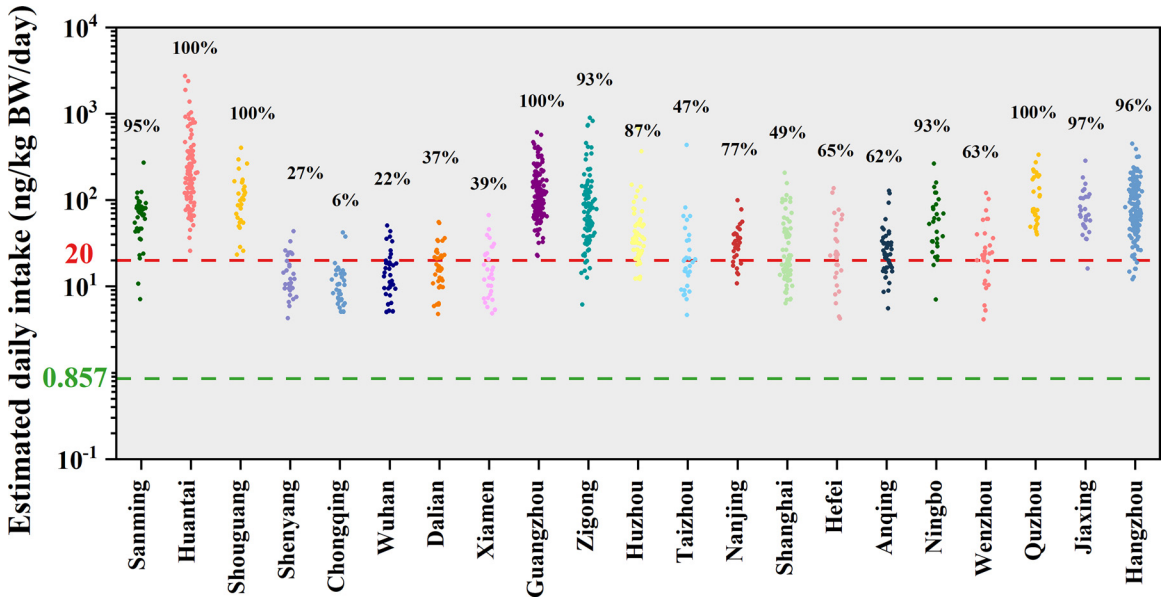


Figure 3. Lactational estimated daily intake (EDI, ng/kg BW per day) of PFOA in <1-month-old infants in different cities, compared with reference values. The red dashed line represents the threshold (20 ng/kg BW per day) from the U.S. Environmental Protection Agency.[67] The green dashed line represents the threshold (0.857 ng/kg BW per day) from European Food Safety Authority.[69] Percentage values are ratio of number of samples exceeding 20 ng/kg BW per day to total sample size in city. All EDIs in each city exceeded 0.857 ng/kg BW per day. All calculated EDI values are provided in Excel Table S3. Note: BW, body weight; PFOA, perfluorooctanoate.

PFO2HxA, and PFO3OA exhibited moderate CL_{renal}, comparable to that of PFBS (median = 3.210 mL/kg BW per day). Overall, PFCAs were preferentially excreted via urine compared with PFSAs with the same perfluoroalkyl chain length, especially 6:2 Cl-PFESA, with the lowest CL_{renal} (0.009 mL/kg BW per day).

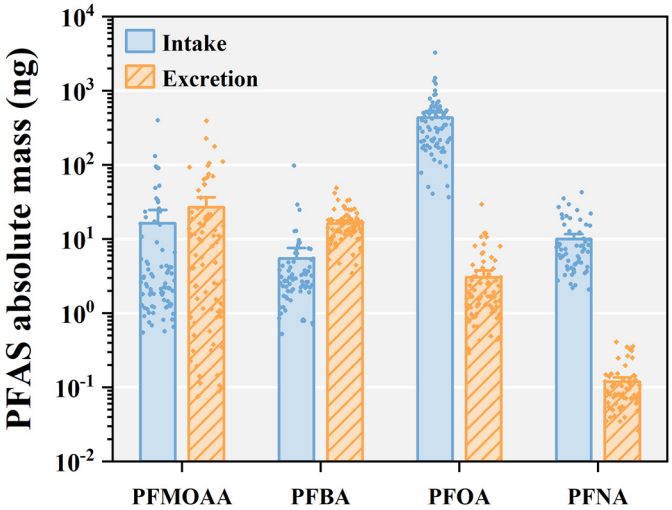


Figure 4. PFAS absolute mass (ng) of daily intake by human milk and excretion by urine in infants. Absolute masses of four PFAS (PFMOAA, PFBA, PFOA, and PFNA) with detection rates >50% in both human milk and urine were calculated. Actual number of paired samples in calculation differed (i.e., $n = 80$, 72, 78, and 59 for PFMOAA, PFBA, PFOA, and PFNA, respectively). Absolute intake was calculated by PFAS concentration (ng/L) in human milk and daily intake rate (510 mL/d).[52] Absolute excretion was calculated by PFAS concentration (ng/L) in urine, clearance rate (48 mL/kg BW per day),[53] and infant weight (kg). Differences between PFAS absolute mass of human milk and urine represent net daily intake for infants. Corresponding raw data are provided in Excel Table S6. Note: PFAS, per- and polyfluoroalkyl substances; PFBA, perfluorobutanoate; PFMOAA, perfluoro-2-methoxyacetic acid; PFNA, perfluorononanoate; PFOA, perfluorooctanoate.

We assumed that renal clearance was the only body elimination pathway for PFAS in infants. Thus, we estimated the half-lives of PFAS in all infants, regardless of sex (Table S13). Although the estimated half-lives of emerging PFECAs were 4–130 times shorter than that of PFOA (median = 6.6 y), 6:2 Cl-PFESA displayed a median half-life of 49 y. Considering continual PFAS intake, we estimated PFAS serum concentrations (in nanograms per milliliter) in infants (<1 year of age) based on measurements in human milk and calculated the CL_{renal} (estimated values are provided in the Excel Table S8). As shown in Figure 5, the model-predicted PFOA level in serum exceeded 100 ng/mL at 6 months of age. The predicted HFPO-TA concentration was consistent with the reported value (Table S14). However, the predicted 6:2 Cl-PFESA level (geometric mean: 6.410 ng/mL) was approximately eight times higher than the measured value (0.847 ng/mL), and the predicted PFMOAA value was approximately one 1/100th that of the measured value in serum (Table S14). The inconsistencies in model prediction for different PFAS are difficult to explain, but may be due, at least in part, to the immaturity of the current prediction model regarding emerging PFAS.

Discussion

The present study was conducted to assess background exposure to PFAS in human milk in various cities in China. Overall, our results showed that PFAS exposure is still widespread in Chinese mothers and that the health implications to neonates are not negligible considering the high detection and abundance of these compounds in human milk, even under increasingly strict worldwide regulations.[7,73] Notably, the concentrations of legacy PFAS, especially PFOA, in breast milk were ~4- to 24-fold higher in the study population (median = 336 ng/L, sample size: $n = 1{,}151$, sampling year: 2020–2021) than in previously studied populations from other countries, such as the United States (median = 13.9 ng/L, $n = 50$, 2019),[31] Czech Republic (median = 44 ng/L, $n = 50$, 2010),[74] Japan (median = 89 ng/L, $n = 30$, 2010),[75] Spain (median = 57.5 ng/L, $n = 10$, 2012),[76] Italy (mean = 60 ng/L,

Table 3. Summary of PFAS renal clearance (mL/kg per day) in infants (present study) and adults (previous studies).

		Infants in present study						Adults in published studies								
Compound		Mean	Min	25th P	Median	75th P	Max	$n=9$[b]	$n=39$[c]	$n=7$[d]	$n=72$[e]	$n=20$[f]	$n=66$[g]	$n=207$[h]	$n=61$[i]	$n=20$[j]
Location	n[a]	Hangzhou/Huantai						Wuhan			Wuhan/Yantai	Shijiazhuang/Handan		Yingcheng		Kyoto
HFPO-TA	24	0.408	0.048	0.108	0.369	0.581	1.206	—	—	—	—	—	—	—	—	—
PFMOAA	79	24.49	0.088	0.978	3.851	17.67	677.5	—	—	—	—	—	—	—	—	—
PFO2HxA	7	7.217	1.084	2.095	2.605	9.348	23.95	—	—	—	—	—	—	—	—	—
PFO3OA	26	3.409	0.054	1.338	2.349	4.324	11.54	—	—	—	—	—	—	—	—	—
PFBA	21	34.61	4.076	15.86	31.80	46.40	93.89	28.70	3.820	10.30	—	—	—	—	—	—
PFPeA	63	25.79	3.305	13.47	21.43	32.95	95.42	—	—	—	—	—	—	—	—	—
PFHpA	33	1.071	0.105	0.540	0.722	1.176	7.679	3.260	17.70	7.980		0.170	0.410	—	—	—
PFOA	78	0.064	0.002	0.031	0.049	0.076	0.259	0.121	0.061	0.079	—	0.140	0.180	0.067[k]	0.070	1.365
PFNA	60	0.082	0.016	0.028	0.047	0.100	0.543	—	0.055	0.074	—	0.200	0.094	—	—	—
PFDA	13	0.037	0.008	0.013	0.022	0.055	0.118	—	0.015	0.031	—	0.047	0.035	—	—	—
PFBS	28	5.537	0.700	2.156	3.210	7.423	27.53	22.20	8.210	174.0	—	—	—	—	—	—
PFHxS	20	0.956	0.052	0.108	0.270	0.781	11.71	6.470	0.012	0.006	—	0.033	0.015	0.023[k]	0.030	—
6:2 Cl-PFESA	33	0.012	0.003	0.006	0.009	0.014	0.053	—	—	—	0.0016	—	—	—	—	—

Note: —, not applicable; Cl-PFESA, chlorinated polyfluorinated ether sulfonates; HFPO-TA, perfluoro-2,5-dimethyl-3,6-dioxanonanoic acid; P, percentile; PFBA, perfluorobutanoate; PFBS, perfluorobutane sulfonate; PFDA, perfluorodecanoate; PFHpA, perfluoroheptanoate; PFHxS, perfluorohexane sulfonate; PFMOAA, perfluoro-2-methoxyacetic acid; PFNA, perfluorononanoate; PFO2HxA, perfluoro(3,5-dioxahexanoic) acid; PFO3OA, perfluoro(3,5,7-trioxaoctanoic) acid; PFOA, perfluorooctanoate; PFOS, perfluorooctane sulfonate; PFPeA, perfluoropentanoate.

[a]Number of paired cord blood–urine samples available for estimating renal clearance.

[b]Background-exposed person ($n=9$).[70]

[c]Fishery employee ($n=39$).[70]

[d]Fishery family ($n=7$).[70]

[e]Predominantly male (58 males/14 females) population ($n=72$).[47]

[f]Young female group ($n=20$).[48]

[g]Male and older female groups ($n=66$).[48]

[h]Occupational workers ($n=207$).[46]

[i]Occupational workers ($n=61$).[71]

[j]Adults ($n=20$).[72]

[k]Values represent average renal clearance in the studied population.

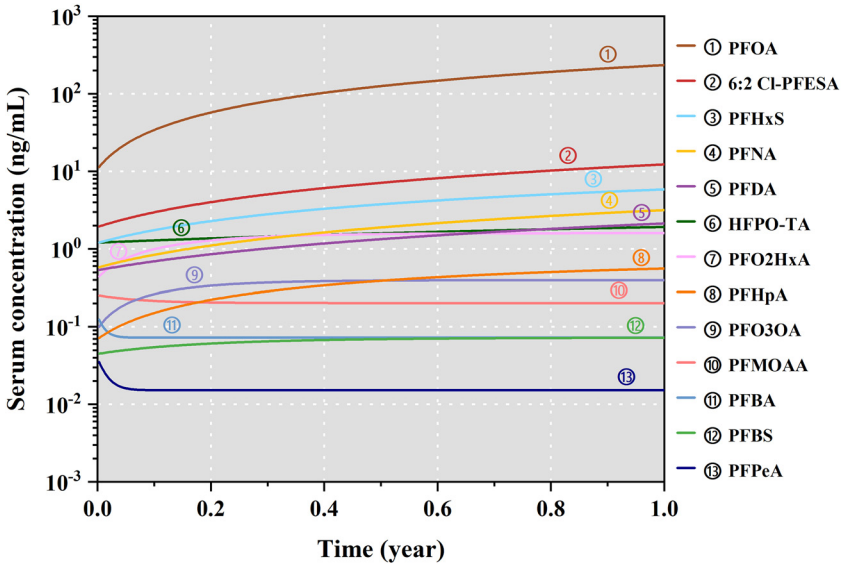


Figure 5. Simulated PFAS serum concentrations (ng/mL) in <1-y-old infants. Simulated values are provided in Excel Table S8. Fitted curves from top to bottom are labeled by ordinal numbers (1–13), responding to the order of PFAS in the right legend. Note: Cl-PFESA, chlorinated polyfluorinated ether sulfonates; HFPO-TA, perfluoro-2,5-dimethyl-3,6-dioxanonanoic acid; PFAS, per- and polyfluoroalkyl substances; PFBA, perfluorobutanoate; PFBS, perfluorobutane sulfonate; PFDA, perfluorodecanoate; PFHpA, perfluoroheptanoate; PFHxS, perfluorohexane sulfonate; PFMOAA, perfluoro-2-methoxyacetic acid; PFNA, perfluorononanoate; PFO2HxA, perfluoro(3,5-dioxahexanoic) acid; PFO3OA, perfluoro(3,5,7-trioxaoctanoic) acid; PFOA, perfluorooctanoate; PFPeA, perfluoropentanoate.

$n = 37$, 2010),[77] and France (mean = 41 ng/L, $n = 61$, 2010–2013),[33] which could be attributed to the ongoing use of PFOA-related products in China.

With the phasing out of PFOA production and usage in North America and Europe, China has become one of the largest fluorochemical manufacturers and consumers in the world,[78] with many fluorochemical plants found across China, such as Shandong Dongyue Chemical Co., Ltd., with a polytetrafluoroethylene (PTFE) production capacity of 45,000 tons/y,[79] and Fujian Sannong Chemical Co., Ltd., with a production capacity of 6,000 tons/y,[79] resulting in an increased risk of exposure to PFOA from environmental discharge into drinking water.[79] Various emerging PFAS used as raw material or intended products in the manufacturing process, including PFMOAA, PFO5DoDA, and 6:2 Cl-PFESA,[2] have been detected in Wilmington residents in North Carolina living downstream of a fluorochemical manufacturing facility.[80] Similar to previous surveys of residents living near fluorochemical plants, PFMOAA, PFO5DoDA, and 6:2 Cl-PFESA were the dominant emerging PFAS detected in the human milk samples (median concentrations = 13.6, 3.42, and 2.63 ng/mL, respectively), accounting for a similar proportion (7.6% in human milk in this study) to that found in a highly exposed population (10.8% in serum).[23] To the best of our knowledge, this study is the first to report on these emerging PFAS in human milk and the first to conduct a comprehensive nationwide (21 cities) survey of baseline PFAS exposure in breastfed infants.

In the present study, there was wide variation in human milk PFAS concentrations in the different cities. The highest median Σ_{22}PFAS concentration was found in Huantai (1,482 ng/L), followed by Shouguang (869.1 ng/L), Quzhou (829.9 ng/L), Guangzhou (824.9 ng/L), and Hangzhou (802.6 ng/L), which were 2–11 times higher than levels in other cities (Table S7). The lack of relevant prenatal data on maternal dietary intake, occupation, and lifestyle habits during pregnancy makes it difficult to explain the discrepancies in the cities studied, although potential pollution sources may offer a partial explanation in some cases where exposure levels are particularly high. Thus, we assessed the composition of PFAS between cities and divided the cities into four clusters (Figure 2A). Cluster 1 consisted of Sanming, Huantai, and Shouguang, all of which contain typical fluorochemical industrial plants.[16,81] This cluster showed significant differences from other cities, especially in the use of PFMOAA and PFO5DoDA (Figure 2B). Cluster 2, consisting of Shenyang, Chongqing, Wuhan, Dalian, and Xiamen, showed PFOA and PFOS as the most dominant PFAS. Cluster 3, consisting of Guangzhou and Zigong, showed a dominant PFOA exposure burden but almost no quantifiable PFO5DoDA in human milk, which may be related to the high PFOA concentration found in drinking water (Zigong: 3,165 ng/L, Guangzhou: 53.4 ng/L).[79] Cluster 4, consisting of 11 cities located in eastern China, showed the highest levels of PFOS and 6:2 Cl-PFESA, similar to the high levels reported in surface water and municipal sewage sludge in the region,[82,83] which may be related to the local electroplating industry.[17,84]

Whether compared with the U.S. EPA-recommended threshold (20 ng/kg BW per day) or the EFSA reference values (0.857 ng/kg BW per day for PFOA and 1.857 ng/kg BW per day for PFOS), the lactational PFOA and PFOS EDI values obtained in this study are of concern, especially regarding their potential health effects, both alone and in combination, on infants. Our estimates for all breastfed infants far exceeded the latest combined tolerable daily intake rates (TDI; 0.63 ng/kg BW per day) for PFOA, PFNA, PFHxS, and PFOS established by the EFSA,[85] and even the minimum estimate was close to 10 times the threshold. Given that birth weight data were missing for five cities, additional sensitivity analysis was conducted for those locations, excluding those with missing birth weight data (Figure S6 and Excel Table S9). The percentages of EDIs exceeding the U.S. EPA-recommended threshold (20 ng/kg BW per day) or EFSA reference values (0.857 ng/kg BW per day for PFOA and 1.857 ng/kg BW per day for PFOS) in each city remained unchanged.

At present, there are insufficient studies to prove that the potential risks of PFAS exposure in infants via breast milk intake reduce the benefits of breastfeeding in terms of growth outcomes. However, based on previous experiments in rodents, exposure to PFAS in early life can have adverse developmental effects, such as growth restriction, obesity, and endocrine disruption.[86,87] Thus, our results suggest that breastfeeding may be a critical

contributor to adverse effects later in childhood. However, we must be careful in interpreting these results because, as infants gain body weight and reduce human milk intake, EDI values decrease with infant age. Although average EDI in male infants was slightly lower than that in female infants (Table S7), the difference was not significant, suggesting that sex did not impact exposure risk during infancy. To date, only trace amounts of PFAS have been reported in infant formula,[74,75,88] in contrast to the high detection and abundance of PFAS in human milk. This suggests that infant formula feeding may be safer than breastfeeding from the perspective of PFAS exposure. Nevertheless, considering the irreplaceable role of human milk in strengthening the immune system and providing balanced nutrients to infants, further epidemiological studies are needed to demonstrate whether breastfeeding with contaminants has adverse health outcomes on newborns.[89]

In the paired samples, the different PFAS composition in human milk and urine could be attributed to their polarity. For example, short-chain PFAS can be readily transferred in the aqueous phase because of their relatively high polarity and water solubility. Human milk, as a complex mixture of endogenous compounds, contains a high lipid content, multiple proteins, vitamins, and antibodies.[88] Therefore, the percentage contribution of PFMOAA and PFBA increased significantly from 2.4% and 1.6% in human milk to 30% and 49% in urine (Figure S4A). In contrast, the proportions of long-chain PFAS, such PFOA, PFOS, and 6:2 Cl-PFESA, were basically constant, accounting for 70%, 9%, and 4% in human milk and 58%, 10%, and 8% in cord blood, respectively. The similar PFAS composition profiles in human milk and cord blood were primarily ascribed to their protein-binding affinity.[90] Comparing daily breast milk intake and urinary excretion, only short-chain PFAS (i.e., PFMOAA and PFBA) were underaccumulated in the body. This result may be interpreted as the original infant body load transferred from cord blood at birth (Figure S4B). However, in the simulated serum concentrations derived from the PFAS clearance rate and cord blood level, PFMOAA was not completely excreted over time (Figure 5). In contrast, previous research has shown that serum PFMOAA is positively correlated with age,[23] suggesting other nonnegligible exposure routes for infants in addition to human milk, such as ingestion of dust and dermal absorption.[30] However, given the difficulty of direct blood collection in infants, we regarded PFAS levels in cord blood to be the same as levels in neonate blood. Although measurements of xenobiotics in cord blood are common, such as estimation of total body burden in newborns,[56] comparison of gestation and lactation exposure,[54] and calculation of placental transport efficiency, simultaneous evaluation of multiple neonatal matrices is recommended to avoid bias from inappropriate sample types.

In this study, we explored PFAS CL_{renal}s and half-lives in infants, especially emerging PFECAs, which have not yet been estimated in humans based on current literature. Overall, the median CL_{renal} values of long-chain PFCAs were 0.049 mL/kg BW per day for PFOA, 0.047 mL/kg BW per day for PFNA, and 0.022 mL/kg BW per day for PFDA, which are much lower than those reported in prior studies (Table 3). Results showed that the estimated median values of PFHxA and PFHxS (0.722 and 0.270 mL/kg BW per day, respectively) were in the range of those reported in adults in prior studies (0.170–17.70 and 0.006–6.470 mL/kg BW per day, respectively).[46,70,71] However, the median renal clearance efficiencies of PFBA and 6:2 Cl-PFESA were slightly higher than those reported previously in adults (Table 3). The reason for these differences is unclear (although participant age, location, diet, and sampling size may be a factor), but in all cases, renal PFAS clearance was consistently ranked in the following order: PFBA > PFHpA > PFOA > PFNA > PFDA. In addition, the corresponding half-life of PFOA was 6.6 y in infants, compared with 2.9 y in low-exposure adults and 8.5 years in high-exposure adults estimated in prior studies.[44] It should be noted that our study used single time-paired whole blood and urine samples and assumed renal clearance as total clearance in the half-life calculation, which may cause an overestimation of half-life. Further, deviation caused by conversion of PFAS levels in whole blood to serum should also be considered. Although previous studies have reported serum-to-whole blood ratios of 1.9–2.3 for most legacy PFAS (e.g., PFOA, PFNA, PFDA, PFHxS, and PFOS),[58] we still found slight differences in these ratios for different PFAS due to their different protein-binding affinities and erythrocyte-binding capacities.[91]

Based on the higher estimated half-lives (3- to 8-fold higher in infants than in adults), our results showed greater accumulation of long-chain PFCAs (>8 carbons) in infants compared with adults,[48] which is unsurprising given the immature metabolic system of newborns. Therefore, the slower excretion of PFAS in infants than in adults provides an opportunity to address the renal clearance and half-lives of these emerging PFECAs, in which oxygen atoms inserted between carbon chains lead to higher water solubility than in legacy PFAS and to faster excretion via urine.[92,93] One of the strengths of our study is that neonatal urine samples were successfully collected with disposable gel absorbent diapers, which made it possible to evaluate PFAS elimination.

We used a one-compartment model to estimate renal clearance and temporal trends of PFAS concentrations in infants. Although this model is simple and easy to implement, it has several limitations: For example, *a*) it cannot fully describe the distribution, metabolism, and excretion processes; and *b*) it does not consider physiological life stage characteristics. Various physiologically based pharmacokinetic (PBPK) models for different PFAS have been published for different age groups,[87,94] including a recent model for sensitive populations (e.g., pregnant women, fetuses, lactating mothers, neonates).[87] However, for some emerging PFAS, model parameters are often lacking. Our data not only provide validation for the PBPK model in infants, but also provide reliable estimates of crucial parameters, such as absorption and excretion factors.

The CL_{renal}s of emerging short-chain PFECAs (PFMOAA, PFO2HxA, and PFO3OA) were lower than those of PFBA and PFPeA but comparable to that of PFBS (3.210 mL/kg BW per day), indicating that the PFECAs were not rapidly eliminated. Also unexpectedly, HFPO-TA was excreted faster than PFOA in infants, in contrast to the higher bioaccumulation potential of HFPO-TA than PFOA reported in studies on wild common carp[16] and animal models.[95] Some bias may exist in the results owing to the large variation in sample size in single PFAS elimination estimates. However, in our study, no other reference data were available to compare PFECA estimates. Although PFECAs showed lower accumulation and faster excretion than PFOA in the studied infants, ongoing exposure from daily human milk intake deserves attention owing to the potential effects on neonatal health. For example, significantly high levels of PFECAs (i.e., HFPO-TA, PFMOAA, PFO4DA, and PFO5DoDA) have been detected in children <1 to 18 years of age.[23] Studies have also found elevated PFECA concentrations in vegetables, eggs, and seafood,[24,96] suggesting exposure risk for consumers. Therefore, epidemiological studies on infant exposure to emerging PFAS are required to determine their associations with growth and developmental outcomes.

Our study showed a marked decrease in clearance rates with increasing carbon chain length for both legacy PFAS and emerging PFECAs, which may be related to their binding affinities to serum albumin[97] and the transport proteins governing reabsorption.[98] For

example, PFOS shows strong interactions with human serum albumin[71] and long-chain PFAS and organic anion transporting protein (Oatp 1a1) show strong interactions in the rat kidney, resulting in lower urinary excretion.[99] However, the above theory fails to explain the presence of 6:2 Cl-PFESA but not PFOS in the urine samples in our study, even though 6:2 Cl-PFESA is longer. This finding is also inconsistent with previous research showing slower urinary elimination of 6:2 Cl-PFESA than PFOS in highly exposed metal plate workers.[47] Therefore, molecular chain length–dependent mechanisms cannot fully explain the discrepancies found in PFAS elimination in different studied populations, and other transport mechanisms are likely acting on PFAS to cross fluid barriers in humans. In addition to the discrepancy in 6:2 Cl-PFESA and PFOS elimination, we also found a 6-fold difference in the estimated half-life of 6:2 Cl-PFESA in infants compared with highly exposed adult populations.[47] Although this difference may be attributable to differences in study population (e.g., age, occupation, and sex composition), dose-dependent renal clearance estimation and ongoing exposure cannot be ruled out, which may result in slower elimination in highly exposed populations.[45] However, despite the uncertainty surrounding the urinary elimination of 6:2 Cl-PFESA and PFOS, their potential health risks to infant development cannot be ignored. Notably, previous toxicological studies have shown that 6:2 Cl-PFESA induces developmental and cardiac toxicity in zebrafish embryos[100] and causes serious liver injury and lipid metabolism disturbance in rodents.[101]

To the best of our knowledge, this study is the first to report on the clearance rate of PFECAs in humans. However, our study has several limitations regarding CL_{renal} and half-life estimates owing to the lack of V_d data for most PFAS. We tentatively used V_d values in rats and mice for the calculation of half-life to ensure estimations were as accurate as possible. However, there were some inevitable uncertainties in the current model based on the differences in toxicokinetic parameters for PFAS among various species. Taking PFOA as an example, previous V_d values have been estimated at 346 and 211 mL/kg BW in male and female rats,[102] 226 and 135 mL/kg BW in male and female mice,[103] 181 and 198 mL/kg BW in male and female cynomolgus monkeys,[104] and 170 mL/kg BW in humans.[59] Thus, we should express caution in discussing the half-lives of PFAS when extrapolating V_d values from animals to humans. As such, additional data on distribution volumes of emerging PFAS are needed to improve the reliability of future CL_{renal} and half-life estimates. In addition, the relatively small number of study subjects decreased the reliability of the PFAS excretion estimates. Furthermore, although we succeeded in obtaining neonatal urine samples, long-chain PFAS levels in infant urine were too low to estimate half-lives. Given the limitations in infant urine collection, we could not obtain fresh urine samples and needed to add $CaCl_2$ salt to release urine from the gel absorbents. The PFAS concentrations in urine were not adjusted for specific gravity, which may interfere with the dilution or concentration of urine. The health risks of PFAS may also be underestimated given that the half-life estimates were taken at a single time point, despite infants being continuously exposed. Therefore, sample collection over an extended period should be considered in future studies to calculate PFAS elimination and determine the correlations between PFAS exposure and infant growth outcomes.

Conclusions

We determined the occurrence and distribution of PFAS in human milk collected from 1,151 lactating women in 21 cities in China and provided baseline data on the potential exposure risk for breastfeeding infants. Our results showed that PFOA, PFOS, and 6:2 Cl-PFESA were the dominant PFAS in breast milk. To the best of our knowledge, this study is the first report on emerging PFECAs in breast milk samples, demonstrating wide distribution in China. Several PFECAs found with high detection frequencies (such as PFMOAA and PFO5DoDA) deserve greater attention. Of concern, the EDIs for PFOA in more than half of the breastfed infants in most studied cities were higher than the recommended threshold, although this should be interpreted with caution given the rapid changes in infant weight and breast milk ingestion. Based on measurements of PFAS in matched cord blood and urine samples, albeit with a small sample size, we determined the PFAS CL_{renal}s and half-lives in infants and predicted PFAS levels in serum during the lactation period, which broadens our understanding of the elimination and accumulation of emerging PFECAs and potential health risks of both legacy and emerging PFAS in infants. However, further studies are needed to explore the association between PFAS exposure and health outcomes and to elucidate the key factors contributing to adverse health effects.

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